25900551 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEPES200 - FUNDAMENTALS OF KINESIOLOGY AND BIOMECHANICS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. How does kinesiology contribute to injury prevention in sports? [A] / [CO1]

   a). By identifying risk factors and improving movement patterns

   b). By developing new sports equipment

   c). By creating new game rules

   d). By improving athletic performance

2 What is the primary action of the rectus abdominis muscle? [K] / [CO3]

   a). Extension of the trunk

   b). Flexion of the trunk

   c). Rotation of the trunk

   d). Stabilization of the pelvis

3. What is the benefit of biomechanical analysis for coaches? [K] / [CO4]

   a). Improved understanding of athlete movement patterns

   b). Enhanced ability to provide feedback on technique

c). Development of more effective training programs
d). All of the above

4. What is one of the applications of kinesiology? [U] / [CO2]

a). Sports and athletics
b). Rehabilitation and physical therapy
c). Health promotion and disease prevention
d). All of the above

5. Which of the following is a benefit of biomechanical analysis in physical education? [K] / [CO4]

a). Improved student performance
b). Enhanced teacher feedback
c). Increased student motivation
d). All of the above

6. What is the insertion of the biceps brachii muscle? [K] / [CO3]

a). Olecranon process of the ulna
b). Radial tuberosity and bicipital aponeurosis
c). Coronoid process of the ulna
d). Lateral epicondyle of the humerus

7. What is the origin of the gluteus medius muscle? [K] / [CO3]

a). Ilium
b). Femur
c). Pelvis
d). Vertebral column

8. Which of the following is a unit of measurement for velocity? [U] / [CO5]

a). Meters per second (m/s)
b). Newtons (N)
c). Joules (J)
d). Watts (W)

9. What is the unit of measurement for acceleration? [A] / [CO6]

a). Meters per second (m/s)
b). Meters per second squared (m/s^2)
c). Newtons (N)
d). Joules (J)

10. How does kinesiology help improve athletic performance? [U] / [CO2]

a). By optimizing training programs and techniques
b). By developing new sports equipment

c). By analyzing movement patterns and mechanics

d). All of the above

[1x10 = 10]

Part B

SHORT ANSWER QUESTIONS

Answer any **10** questions

Each question carries **2** marks

11. Provide an example of flexion and extension in the human body. [K] / [CO1, CO2]

12. What is the primary role of biomechanics in sports? [U] / [CO4]

13. Describe the origin of the trapezius muscle. [K] / [CO3]

14. Describe the origin of the adductor longus muscle. [U] / [CO3]

15. What is torque in biomechanics? [K] / [CO5]

16. What is abduction in the context of human movement? [K] / [CO1, CO2]

17. State the main functions of the hamstrings. [K] / [CO4]

18. Why is understanding force and momentum important in sports? [U] / [CO6]

19. Name two muscles of the rotator cuff. [K] / [CO3]

20. What is the unit of measurement for velocity? [U] / [CO5]

21. What is circumduction in the context of human movement? [K] / [CO1, CO2]

22. Mention the origin and insertion of Pectoralis Major. [K] / [CO3, CO4]

[2x10 = 20]

Part C

SHORT ESSAY QUESTION

Answer any **4** questions

Each question carries **5** marks

23. Discuss the laws of motion and their application. [A] / [CO5]

24. Explain the historical development of kinesiology from ancient times to the modern era. [U] / [CO1]

25. List the muscles forming the erector spinae group and describe their anatomical arrangement. [K] / [CO3]

26. Discuss the importance of supination and pronation movements in daily activities. [U] / [CO2]

27. Discuss the importance of the hip flexors and gastrocnemius muscles in daily activities. [U] / [CO3]

28. Describe the different types of joints in the human body. [K] / [CO5]

[5x4 = 20]